MINI PROJECT · LECTURE 1: FOUNDATIONS OF LLMs

MiniGPT: Sequence Reversal

with SFT + GRPO Fine-tuning

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OVERVIEW

Problem & Two-Stage Pipeline

STAGE 1 — SFT

Reverse a sequence

```
Input  <BOS> h e l l o <SEP>
Target o l l e h <EOS>
```

Length 2–10 · 50k samples
Cross-entropy on shifted target

STAGE 2 — RL (GRPO)

Reverse + skip 'b'

```
Input  <BOS> a b c d e <SEP>
Target e d c a <EOS>
```

Length 2–6 · 15k steps
GRPO + KL penalty + curriculum

Why two stages? Skip-'b' is a constraint that's hard to express through supervised loss — RL is well-suited for it.

MiniGPT Architecture

Hyperparameters

Param	Value
d_model	256
nhead	8
num_layers	4
dim_ff	512
dropout	0.1
vocab_size	30
Total params	**~2.1M**

Forward pass

Token Embedding

+ Sinusoidal PE

TransformerEncoder × 4

Linear → vocab logits

Note: uses *TransformerEncoder* + causal mask (torch.triu) — equivalent to a Decoder.

Vocabulary (30): <PAD> · <BOS> · <SEP> · <EOS> + a–z (26 lowercase letters)

STAGE 1

SFT — Supervised Fine-Tuning

Sequence format (shift-by-one)

```
x:  <BOS>  h  e  l  l  o  <SEP>  o  l  l  e  h
y:  h    e  l  l  o  <SEP>  o  l  l  e  h  <EOS>
```

Training config

Optimizer: **AdamW (lr 3e-4)**
Batch: **64**
Epochs: **40**
Loss: **Cross-entropy (ignore PAD)**
Scheduler: **ReduceLROnPlateau**
Clip: **max_norm = 1.0**

SFT can reverse, but doesn't skip 'b'

Input	SFT output	Skip-b target
tesbt	tbset	tset
abcde	edcba	edca
qwer	rewq	rewq

Why RL? GRPO Algorithm

Skip-'b' constraint → no labelled supervised data; reward signal is the natural way to specify it.

1 Sample

Pick prompt; run $g = 4$ stochastic rollouts

2 Reward

Compute $r_1 \ldots r_4$ for each rollout

3 Advantage

$A_i = (r_i - \mu) / (\sigma + \varepsilon)$ within group

4 Update

$-E[\log \pi \cdot A] + KL \cdot \lambda - H \cdot \beta$

Key idea — GRPO uses **the group of rollouts as its own baseline**, so we **don't need a separate value network** (unlike PPO). *Lower-variance gradients than vanilla REINFORCE, simpler to implement than actor-critic.*

Reward Function Design

REWARDS (+)

Condition	Value
Exact match with target	**+5.0**
Positional match (per pos.)	+0.2 × ratio
Character coverage (multiset)	+0.1 × ratio

Design choice: *dense rewards (positional + coverage) reduce gradient variance vs sparse exact-match alone.*

PENALTIES (−)

Condition	Value
'b' in output	**−1.0 × count**
Length mismatch	−0.1 × |Δ len|
No <EOS> generated	−0.5
<PAD> in output	−2.0 × count
<SEP> / <BOS> in output	−2.0 × count

Anti-hacking: *PAD/SEP/BOS penalties prevent the policy from emitting special tokens to game the reward.*

+ KL penalty ($\lambda = 0.2$) · **+ Entropy bonus** ($\beta = 0.005$) · **+ Curriculum** (max_len 3 → 6)

Results — Headline Numbers

IN-DIST · SFT	IN-DIST · RL	OOD · SFT	OOD · RL
22.8%	**100%**	**9.4%**	**14.4%**
exact_match	*exact_match*	*exact_match · len 7-15*	*exact_match · len 7-15*

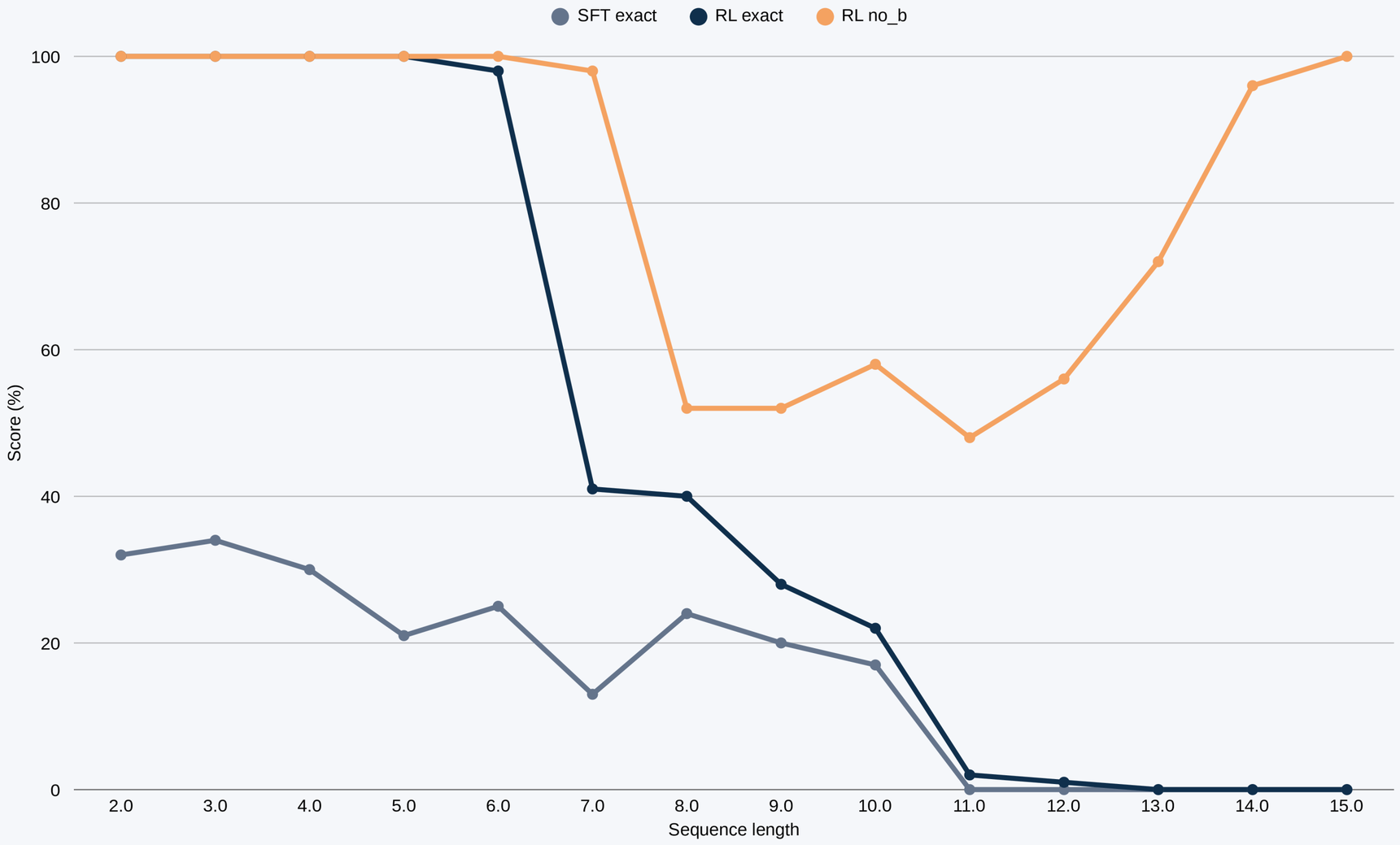
no_b_rate (skip behaviour only)

	SFT	RL	Δ
In-dist (len 2–6)	22.8%	**100%**	**+77.2**
OOD (len 7–15)	22.4%	**69.0%**	**+46.6**

Insight: *skip-'b' behaviour transfers to OOD lengths much better than reversal accuracy does.*

GENERALIZATION

Per-Length Breakdown



In-dist (2–6)

RL: **98–100%**
exact_match
Trained range — works perfectly

OOD (7–15)

Exact drops fast
→ positional encoding
not trained on long ctx
no_b_rate stays high
→ skip behaviour learned
as a transferable rule

Edge Cases — What Works, What Fails

Works ✓	Fails ✗	Why?
bbbbb → ∅ *(all-b → empty)* **qwerty → ytrewq** *(no-b reversed cleanly)* **tesbt → tset** **banana → anana** *(skip-b in reversal)*	**b → j** *(should be empty)* **basketball → llabteksa** *(target: llateksa, 10 chars > train range)*	**Single 'b':** min_len=2 → never seen in training; SFT prior expects ≥1 letter before <EOS> **Long inputs:** Sinusoidal PE not trained on len > 6 in RL

TAKEAWAYS

Discussion & Future Work

GRPO works

100% in-dist exact_match at step 9.4k / 15k — a 2.1M-param model can learn structured constraints fast.

KL prevents forgetting

Model still reverses correctly on no-b inputs (qwer → rewq) — KL = 0.2 keeps reversal skill intact.

Generalization gap

Skip-'b' rule transfers (no_b high on OOD), but reversal accuracy collapses with sequence length.

Future Work

- → Extend RL training range to length 2–12 to address OOD generalization
- → Increase grpo_g or batch size for lower gradient variance
- → Add reward shaping for empty-output cases (e.g. all-b, single-b)
- → Try learnable / RoPE positional encoding for better length extrapolation

Q & A

Thank you · We're happy to take questions

Code & report: MiniGPT — Reverse + Skip-b

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